

Internal Charging Issues Met During Electric Orbit Raising and in GPS-Like MEO Environment

T. Paulmier¹, D. Lazaro, B. Dirassen, R. Rey, J.-C. Matéo-Velez², and D. Payan¹

Abstract—This paper was dedicated to the experimental characterization and numerical assessment of the possible risk of charging during the electric orbit raising (EOR). Different elements [printed circuit board (PCB), ethylene tetrafluoroethylene wires (ETFE)] considered at risk in representative electron environment have been tested to assess their charging behavior and any risks of electrostatic discharges in this environment. This task has been performed in the GEODUR irradiation test facility installed at Office National d'Etudes et de Recherches Aéronautiques (Toulouse, France) and equipped with a 2.5-MeV electron accelerator. The samples have been irradiated with an electron spectrum representative of the maximum spectrum (in regard of electron flux) met during the EOR behind 1-mm shielding. This experimental study revealed high charging kinetics on the different elements over successive cycles that led to the initiation of several electrostatic discharges on PCB, located on metallic floating parts. This paper demonstrated the risk of high charging and electrostatic discharge (ESD) risks during EOR on sensitive components. These discharges could affect the electronics operation and damage the insulators. The final step of this paper was to perform numerical simulation for internal charging assessment in global positioning system (GPS)-type orbit on different space used materials with a dedicated tool that takes into account the temporal evolution of high energy electron fluxes. Then we demonstrated a strong evolution of internal charging with the radiation environment, which has been ascribed to the influence of radiation-induced conductivity (RIC). Strong discrepancies from one material to the other have been observed later on the temporal evolution of the surface potential.

Index Terms—Dielectric materials, electric orbit raising (EOR), electrical conductivity, electron radiation effects, internal charging, materials reliability, polymers.

I. INTRODUCTION

ELECTRIC propulsion is used increasingly by different operators to reach geostationary orbit (GEO) so as to reduce propellant mass and then either increase the payload or choose costless launches. The counterpart of this new technology is that the spacecraft, before its operation in GEO, spend several months within the radiation belts during orbit raising. Radiation belt electrons can go through the

outer shielding of the spacecraft, irradiate sensitive insulating elements present in cables, printed circuit board (PCB), connectors, and prompt significant internal charging of these elements. The high induced electric fields can then initiate electrostatic discharge (ESD) resulting in permanent damage to the dielectrics, component failure, and false commands. Internal charging has been proven to be responsible for several anomalies on spacecraft [1]–[3]. The spacecraft might then be strongly disturbed during orbit raising. Different inner components could be at risk regarding internal charging. Internal charging is indeed steered by complex physical mechanisms induced by the large range in energy for the incident electrons that go through the outer skin of the spacecraft to get implanted on thick dielectric materials. High energy electrons can indeed induce both ionization and implantation within the components that lead to specific electric field configuration built up by charge implantation. Ionization process steers at a great level the effective electric conductivity of dielectric materials: we speak about radiation-induced conductivity (RIC) [4]–[6]. RIC can strongly affect charge transport even for low electron fluxes encountered in the spacecraft. The complexity of internal charging lies on the geometry of the irradiated elements and the grounding configuration that can enhance significantly the induced electric field. Moreover, the use of floating metallic parts can be encountered on spacecraft on some components which could further the initiation of electrostatic discharges and consequent anomalies [7]–[9]. Internal charging process may strongly depend on the electron environment met by the spacecraft during its overall mission, as this is the case for surface charging [10]. For internal charging, worst irradiation configuration might be different from one system to the other, depending on its geometry, grounding scheme, and material properties. The objective of this paper was, therefore, to extract and reproduce experimentally the worst case spectra met during electric orbit raising (EOR) and their evolution as a function of time and to measure the charging behavior and risk of electrostatic discharges on different sensitive components such as PCB, and ethylene tetrafluoroethylene (ETFE) wires. The second step of this paper was to assess numerically the charging behavior of different polymer materials in global positioning system (GPS)-like orbit and demonstrate the high sensitivity of these materials to the environment temporal evolution.

II. ELECTRON SPECTRUM AND FLUX DURING EOR

Evolution of electron fluxes during EOR using super-synchronous transfer orbit (SSTO) orbit has been assessed at Office National d'Etudes et de Recherches Aéronautiques (ONERA) [11]:

Manuscript received October 31, 2018; revised May 20, 2019; accepted June 7, 2019. Date of publication July 1, 2019; date of current version August 9, 2019. This work was supported by Office National d'Etudes et de Recherches Aéronautiques in the frame of the ORESTE Research Project. The review of this paper was arranged by Senior Editor H. B. Garrett. (Corresponding author: T. Paulmier.)

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Digital Object Identifier 10.1109/TPS.2019.2922256

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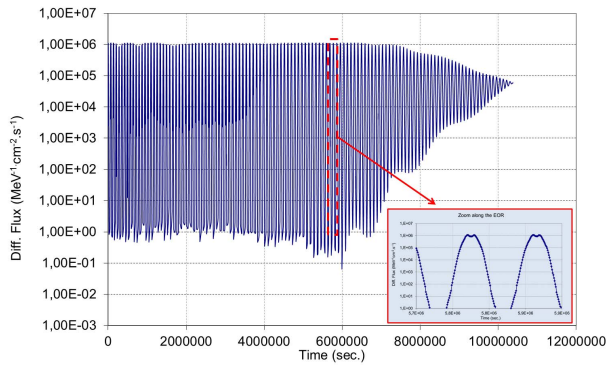


Fig. 1. Evolution of electron flux at 2 MeV along the overall duration of the EOR for the SSTO mission and zoomed-in view on a period for which electron flux is at its maximum value (assessed with the AE8max model).

SSTO Trajectory: 250 km $\times$ 60 000 km, representative of the initial orbit of Eutelsat 115 West B, ABS 3A launched by Falcon 9.

The electron environments have been extracted using the AE8max model [12]. The AE8 model (AE8 is an average model with two versions AE8MAX and AE8MIN, respectively, for maximum and minimum solar conditions) is used in this paper since the electron environment is not known accurately along the overall trajectory of the EOR, especially at low altitude. In Fig. 1, we can observe a strong evolution of the electron flux along the SSTO orbit with maximum levels that could lead to high charging risks within the spacecraft. We can observe that the spacecraft should be submitted to the strong variation of the electron flux by about six decades in about 14 h from the first moments of the mission. The SSTO radiation conditions are indeed defined by the trajectory during EOR and therefore by the evolution of altitudes all along the trajectory. During the SSTO mission, the spacecraft goes through the radiation belts (back and forth) yielding to a significant variation of the received electron flux. These fluxes have been simulated along trajectories with the AE8 model running on the OMERE software (<http://www.trad.fr/OMERE-Software.html>). This evolution on the electron flux might also lead to specific evolution of the charging dynamics of the irradiated systems, depending on the electric properties of the materials. Therefore, Fig. 1 provides information on the dynamics of the electron environment during the spacecraft transfer. For internal charging, we need to assess the electron spectrum affecting the inner elements behind a specific shielding that depends on the configuration of the spacecraft. The higher charging kinetics would be met when the electron fluxes for the highest energies behind shielding present a maximum level. Later, it has been decided (see inset in Figs. 1 and 5 for a better view) to test components at moments during which the spacecraft has to cope with maximum electron fluxes. For this, we assessed the electron spectrum met when the electron flux at 2 MeV is at its maximum level. In our facility, we cannot test the samples with a duration corresponding to the overall orbit and variations. We had to choose a window (see Fig. 1) within the EOR orbit to demonstrate a potential risk of charge and discharge.

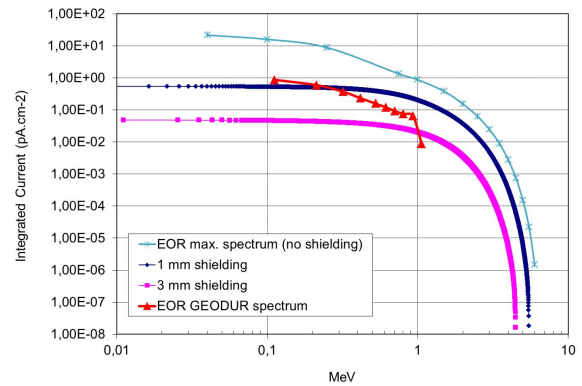


Fig. 2. EOR electron spectrum assessed for the SSTO mission without shielding (input spectrum for OMERE calculations) and behind 1- and 3-mm shielding (and compared with the GEODUR spectrum adjusted for EOR).

Electron spectra along the orbit have been calculated behind 1- and 3-mm Al shielding (see Fig. 2) using the OMERE software. Electron transport behind 1- and 3-mm Al shielding has been assessed in the OMERE tool using the Transport Electron Analytical model (TEAM) developed at ONERA and based on empirical laws devised by Subba Rao [13] and constructed from experimental data. This tool calculates the transmitted differential flux behind an aluminum shielding with a given thickness. The geometrical configuration is a finite thickness slab.

This paper has been prepared for the EOR SSTO mission. We can notice in Fig. 2 that the integrated electron flux (meaning for one energy E , integration of differential currents for energies higher than E) for 3-mm shielding exceeds the usual maximum allowed electron flux (0.02 pA cm^{-2}) adopted in European Cooperation for Space Standardization (ECSS) standards [14].

For the assessment of internal charging risks and detection of discharges on the irradiated components, it is important to test experimentally the systems considered at risk, in representative irradiation conditions regarding both charging flux levels and electron flux dynamics. This is the objective of this paper.

III. GEODUR IRRADIATION FACILITY

The GEODUR facility, funded by Centre National d'Etudes Spatiales and installed at ONERA (Toulouse, France), has been used for the experimental characterization of internal charging. GEODUR is a radiation test facility allowing the study of spacecraft internal and surface charging, evaluation of Radiation-induced conductivity of thick materials and sample radiative aging using 400 keV to 2.5 MeV monoenergetic electrons. It is equipped with a 2.5-MeV Van de Graaff electron accelerator and a double scattering system for the production of a distributed electron spectrum in the energy range (200 keV–1 MeV). It is instrumented with a contactless electrostatic probe and current measurement systems for the characterization of internal charging behavior of space elements. The temperature of the sample holder can be controlled in the range ($-180 \text{ }^{\circ}\text{C}$, $+250 \text{ }^{\circ}\text{C}$) allowing to

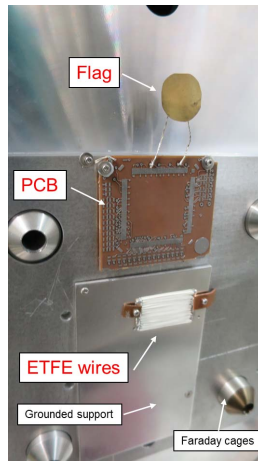


Fig. 3. View of the PCB and ETFE wires tested in the GEODUR facility.

reproduce the temperature variations of materials on the flight. A pumping system allows experiments at a vacuum of around 10^{-6} hPa. Four Faraday cages are placed around the sample support (as seen in Fig. 3) to control the electron incident current. This facility is designed to reproduce a distributed spectrum close to the EOR electron spectrum (see Fig. 2). This electron spectrum is adjustable in flux so as to get as close as possible to the required environment spectrum, such as worst-case spectra. This facility has been used in this paper, especially for the characterization of materials and systems in representative electron irradiation behind shielding (in an energy range between 200 keV and 1 MeV).

IV. CHARGING BEHAVIOR OF A SPACE USED PRINTED CIRCUIT BOARD AND ETFE WIRES

Two different components have been tested in SSTO irradiation conditions: PCB and ETFE wires (see Fig. 3). The tested space used PCB is a multilayer PCB with internal metallic planes connected to ground in our experiments. This PCB is composed of metallic tracks at the surface, which were left at floating potential during the experiments. The different metallic tracks at the surface of the PCB are directly exposed to electron irradiation (the PCB is not coated with any varnish). The different tracks are made with copper. About ten floating tracks have been connected to a circular metallic brass flag which allows measuring the surface potential built upon these tracks during irradiation, using a noncontact Kelvin probe (TREK 3455ET), that moves along the sample and scans the flag and the free surface of the PCB. The flag does not influence charge accumulation on the PCB since it is screened from the irradiation by a metallic shield fixed on the support of the Kelvin probe (see Fig. 4). The ETFE sample is composed of a sheet of several 5-cm long pieces of wire so as to produce a significant surface area for the electric potential measurements. These samples have been tested with a representative electron spectrum behind 1-mm shielding. We tried to reproduce temporal variations of the flux (see Fig. 5) with irradiation for 5 h followed by 19 h relaxation. This dynamics is a rough simulation of the dynamics on the flight since it is not possible to change the electron flux

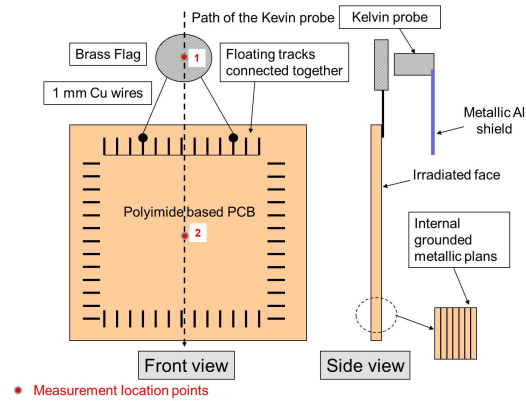


Fig. 4. Basic scheme of the tested PCB sample and the experimental set-up.

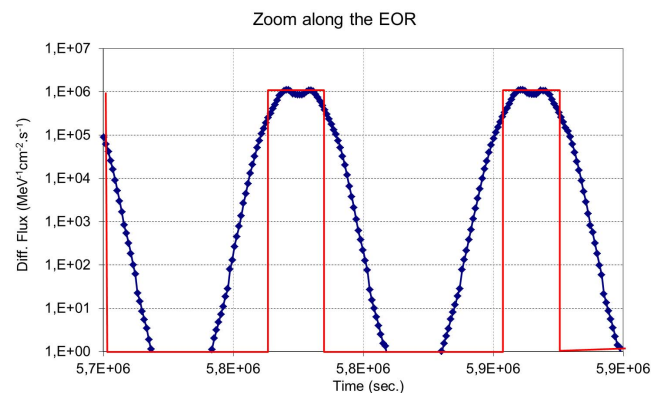


Fig. 5. Experimental simulation (red curve) of the electron flux dynamics in SSTO EOR conditions.

continuously during the irradiation process. The duration of the irradiation step has been defined as the time for which the electron flux is higher than 30% of its maximum value at 2 MeV (Fig. 5). Below this level, we can indeed consider that the charging level is low and this phase would then be linked to a relaxation process. Four irradiation cycles have been applied to simulate 4 d on the flight (a cycle is one irradiation period followed by a relaxation phase). These four cycles are later composed of the following phases.

- 1) Cycle 1: 5-h irradiation, 19-h relaxation
- 2) Cycle 2: 5-h irradiation, 19-h relaxation
- 3) Cycle 3: 5-h irradiation, 19-h relaxation
- 4) Cycle 4: 5-h irradiation, 19-h relaxation

Figs. 6–9 show the evolution of potentials measured on PCB (center of the sample), PCB flag, and ETFE wires during the four irradiation cycles at 20 °C (temperature is supposed to be constant, close to 20 °C within the spacecraft). The electron flux is the same during irradiation for the four cycles. We adjusted the electron spectrum and flux for the PCB to be irradiated by a representative EOR spectrum behind 1-mm shielding (see Fig. 2). We can observe strong and steady charging kinetics for both components. This behavior can be explained by the low intrinsic and radiation-induced conductivity (RIC) for both materials [4] used on PCB and wires (polyimide-based material for PCB and ETFE for wires).

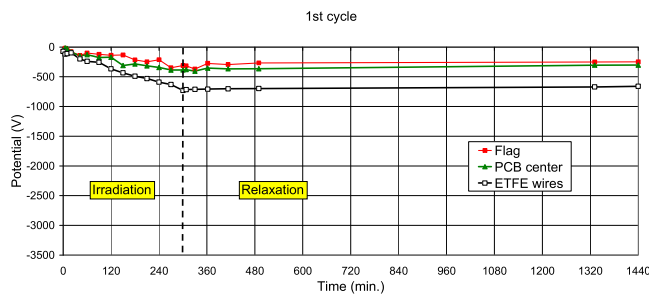


Fig. 6. Evolution of surface potential measured on PCB, PCB flag, and ETFE wires during the first cycle.

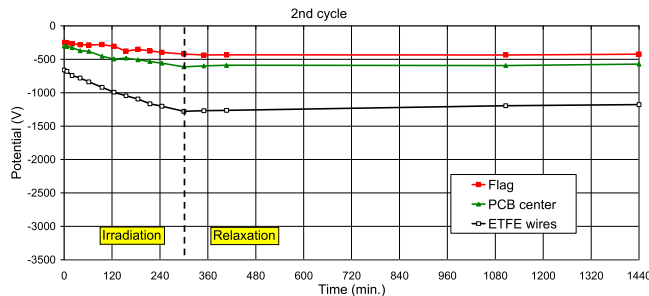


Fig. 7. Evolution of surface potential measured on PCB, PCB flag, and ETFE wires during the second cycle.

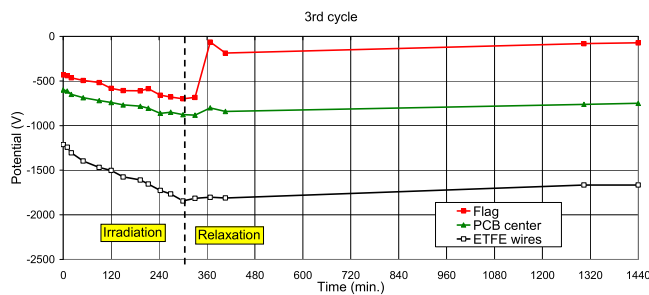


Fig. 8. Evolution of surface potential measured on PCB, PCB flag, and ETFE wires during the third cycle.

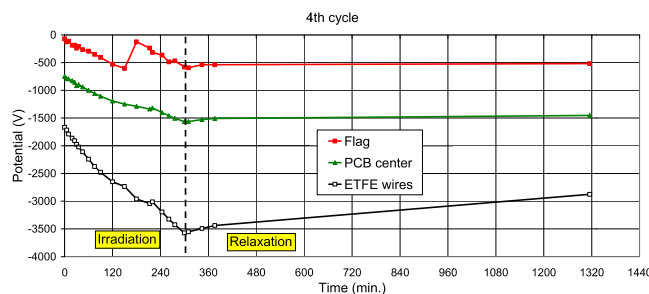


Fig. 9. Evolution of surface potential measured on PCB, PCB flag, and ETFE wires during the fourth cycle.

This low conductivity is highlighted by the low relaxation kinetics observed after each irradiation step. Bulk conductivity for these materials is indeed lower than $10^{-19} \Omega^{-1} \cdot \text{m}^{-1}$ and RIC for the applied flux is $2.10^{-15} \Omega^{-1} \cdot \text{m}^{-1}$ for polyimide [21] (ETFE cables do not present any significant RIC [20]). Later, charges tend to get accumulated in the inner

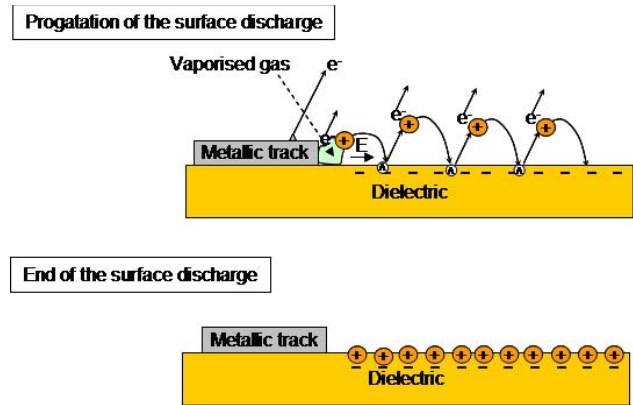


Fig. 10. Surface metal—dielectric discharge with a metallic floating track under electron irradiation.

material up to the initiation of electrostatic discharge. Two discharges have been detected on PCB during the third and the fourth cycle.

A dramatic potential drop is indeed observed on the PCB flag after 255 min for the third cycle and after 120 min irradiation for the fourth cycle: this steep decline is the signature of an electrostatic discharge on the floating tracks of the PCB. Electrostatic discharges produced at the surface induce the creation of plasma due to local vaporization of the materials initiated by the discharge. Ions created in the plasma come back to the negatively charged sample which gets then electrically neutralized around the initiation point of the discharge [15]. The overall process (also described in Fig. 10) is described as follows: the potential at the surface of the floating metallic tracks is steered by the underlying dielectrics that store the incident electrons in the bulk. The metallic tracks are not perfect and present microscopic tips that drastically enhances the electric field at these points and induces a significant rise of the charge current densities and then the emission of electrons from the metal (we speak about field emission) at an apparent electric field value much lower than the one meeting for free dielectric surfaces (without any floating metal). The high current density leads to thermal heat-up of the metal and to a local fusion of this material. The potential of the floating metal can then go back to zero. This process generates then a transverse electric field (between metal and dielectrics) and a vaporized gas. A neutral atom in the gas phase can then get ionized and separated into an ion and an electron. The electron is collected by the surrounding ground while the positive ion drifts back to the negative dielectric surface and compensates this negative charge. The deposited ion can induce the desorption of a second atom that, in turn, gets ionized and follow the same process on the neighboring site of the dielectric until the total annealing of the negative charges at the surface of the dielectrics.

In Fig. 8, we can observe a rise of the absolute surface potential after the discharge during the relaxation process (at 360 min). The possible explanation is the following [15]: after discharge, a positive charge layer is generated at the surface of the dielectrics by electron loss from the discharge and ion deposition (see Fig. 10). The electron previously deeply

trapped beforehand tends to drift to the surface under the effect of inverted electric field induced by the positive charges. This electron drift induced an increase of the surface potential (since the effective capacitance created between the charge layer and the ground is reduced). This process has also been observed when irradiated materials are discharged in air and placed again under vacuum (we observed a rise of the surface potential) (see [16]). Therefore, this experiment demonstrates the risk of using floating metallic unused tracks on PCB within the spacecraft. These elements used on spacecraft might be prone to be submitted to frequent electrostatic discharges during EOR. This is, however, important to mention that 1-mm Al-equivalent shielding is close to a worst-case scenario: actually, in regard of spacecraft construction, Al-equivalent shielding might strongly vary with a minimum Al-equivalent shielding of 3 mil (76 μm) [22]. NASA guideline suggests 110 mil (2.8 mm) of Al to limit internal charging on GEO spacecrafts [22]. In EOR, however, incident electron fluxes can be higher than in GEO and this level of Al-equivalent shielding should not be sufficient to prevent all internal electrostatic discharges, as demonstrated in this paper.

Using a 1-mm shielding configuration is similar to applying an acceleration factor. However, we should still observe a steady but slower rise of the surface potential, up to the initiation of electrostatic discharges after 30 or 40 d on the flight during the EOR (which last for 120 d). Finally, ETFE could be subjected to dielectric breakdown for electric fields close to the one reached in this experiment. In EOR, we could have to cope with dielectric failures of insulating material and to the initiation of high leakage currents that could be detrimental to the onboard equipment.

V. NUMERICAL PREDICTION OF POLYMERS CHARGING BEHAVIOR IN MEO ENVIRONMENT

For medium earth orbit (MEO), the issue regarding internal charging processes is quite similar to the one encountered in EOR: the dynamics of the electron environment is very strong leading to a steep variation of the electron current impacting the spacecraft [17] and, therefore, to possible strong evolution on the surface potential and electric field induced by the irradiation on the onboard materials.

The final step of this paper was, therefore, to perform numerical simulation for internal charging assessment under dynamic electron environment. The numerical simulations have been performed with a dedicated tool that takes into account the temporal evolution of high energy electron flux and the effect of radiation-induced conductivity and radiation history effect on the electric properties. We implemented as input data the electron spectrum and its temporal dynamics measured with the SURF spectrometer installed on the GIOVE-A spacecraft [17].

A dedicated physical and numerical model, called Resistance/Capacitance (RC) model, has been developed to describe the evolution of surface potential and RIC of space-used materials irradiated with high-energy electrons in representative charging conditions. This model is based on former studies and models devised by Rose [18] and Fowler [19].

In this physical model, the irradiated sample is modeled as a system of capacitance and electrical resistance in parallel. From this simple model, the surface potential can be calculated with the relation

$$\frac{dV}{dt} = \frac{Lj(t) \cdot (1 - \eta - \beta) - e(\mu_n n + \mu_p p)V}{\varepsilon} \quad (1)$$

in which L is the sample thickness, $J(t)$ is the incident charging flux that varies over time, η is the secondary electron emission yield, β is the surface leakage parameter, V is the surface potential, e is the elementary charge, ε is the material permittivity, n is the density of free electrons, and p is the density of free holes (vacancies of electrons).

The density of free charges can be assessed with the equations of the RC model

$$\frac{dn}{dt} = g(t) - \alpha_1 n(p_{t1} + p_{t2}) - \frac{n}{\tau_{n1}} - \frac{n}{\tau_{n2}} + \frac{n_{t1}}{\tau_{nt1}} + \frac{n_{t2}}{\tau_{nt2}} \quad (2)$$

$$\frac{dp}{dt} = g(t) - \alpha_2 (n_{t1} + n_{t2})p - \frac{p}{\tau_{p1}} - \frac{p}{\tau_{p2}} + \frac{p_{t1}}{\tau_{pt1}} + \frac{p_{t2}}{\tau_{pt2}} \quad (3)$$

$$\frac{dn_{t1}}{dt} = \frac{n}{\tau_{n1}} - \frac{n_{t1}}{\tau_{nt1}} - \alpha_2 p n_{t1} \quad (4)$$

$$\frac{dn_{t2}}{dt} = \frac{n}{\tau_{n2}} - \frac{n_{t2}}{\tau_{nt2}} - \alpha_2 p n_{t2} \quad (5)$$

$$\frac{dp_{t1}}{dt} = \frac{p}{\tau_{p1}} - \frac{p_{t1}}{\tau_{pt1}} - \alpha_1 n p_{t1} \quad (6)$$

$$\frac{dp_{t2}}{dt} = \frac{p}{\tau_{p2}} - \frac{p_{t2}}{\tau_{pt2}} - \alpha_1 n p_{t2} \quad (7)$$

for which $g(t)$ is the electron/hole pairs generation rate that varies as a function of time, α_1 and α_2 are the recombination rates, n , n_{t1} and n_{t2} are densities of free and trapped electrons, p , p_{t1} and p_{t2} are densities of free and trapped holes, τ_{n1} , τ_{n2} , τ_{p1} , and τ_{p2} are trapping time for free electrons and holes, τ_{nt1} , τ_{nt2} , τ_{pt1} , and τ_{pt2} detrapping time for trapped electrons and holes.

Data have been kindly supplied by the Surrey Space Center (U.K.) which developed the SURF spectrometer. GIOVE-A is a GPS-type MEO spacecraft of altitude 23 300 km and inclination 56° . SURF on Giove-A measures deposited currents in three stacked collector plates [17]. Data collected from January 1, 2010 to June 30, 2010, have been extracted for the simulation. Extracted data correspond to electronic current behind 500- μm thick shielding. We can compute two different currents, concerning the involved energies: the charging current [corresponding to electrons that get implanted in the 500- μm thick materials and contributed, therefore, to its charging process—corresponds to $j(t)$ in the model] and the ionizing current [electrons that go through the material and induce ionization and, therefore, RIC—corresponds to $g(t)$ in the model]. Temporal variations of the charging and ionizing currents are shown in Figs. 11 and 12. These data have been used for the RC model to assess the evolution of the surface potential built upon three different space used dielectric materials: polyimide 500- μm thick Kapton, 500- μm thick polyether ether ketone (PEEK), and 100- μm thick ETFE. Thicknesses have been chosen to be representative of those

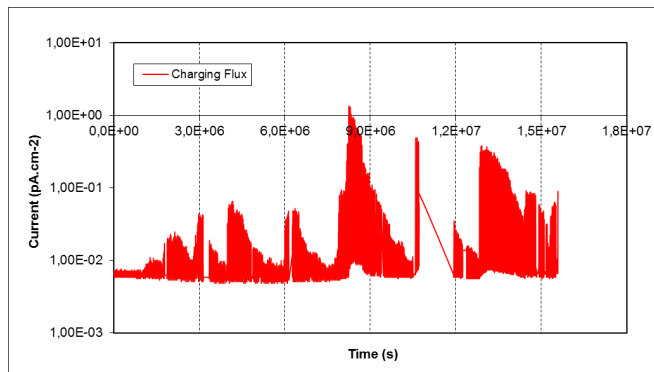


Fig. 11. Temporal evolution of the absolute charging electron current assessed from the SURF data between January 1, 2010 and June 30, 2010.

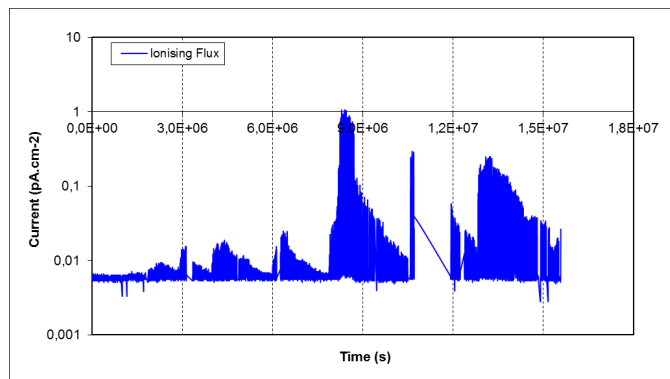


Fig. 12. Temporal evolution of the absolute ionizing electron current assessed from the SURF data between January 1, 2010 and June 30, 2010.

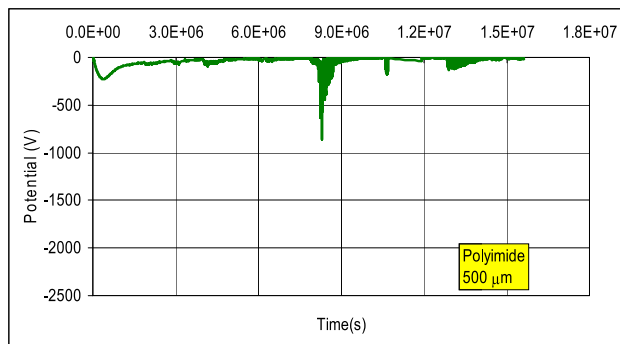


Fig. 13. Evolution of surface potential assessed on polyimide as a function of time in GPS-like orbit behind 500- μ m shielding.

used within the spacecraft. As demonstrated in previous studies [20], [4], these three materials present quite different electric properties with significant RIC for Kapton and PEEK and a low RIC for ETFE. Moreover, if PEEK presents a constant RIC as a function of radiation dose, Kapton, on the contrary, tends to get more and more conductive along with the irradiation: its RIC rises with the increasing radiation dose. The RC model developed at ONERA takes into account this radiation history effect. The charging behavior of these three materials under dynamic electron environment is drastically different as shown in Figs. 13–15.

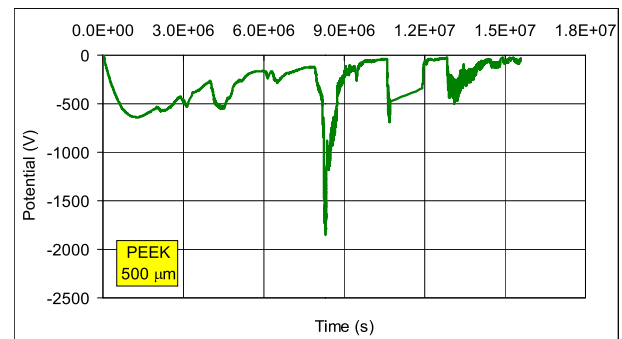


Fig. 14. Evolution of surface potential assessed on PEEK as a function of time in GPS-like orbit behind 500- μ m shielding.

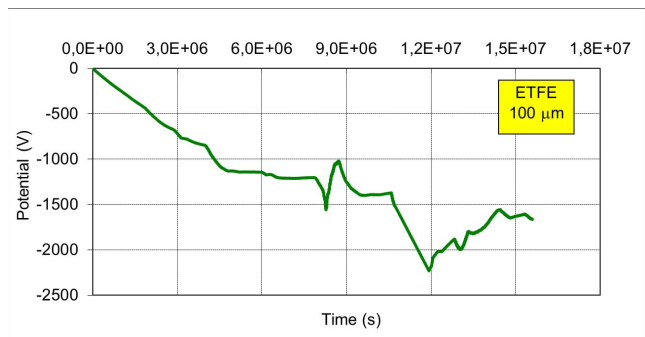


Fig. 15. Evolution of surface potential assessed on ETFE as a function of time in GPS-like orbit behind 500- μ m shielding.

Later, we demonstrated a strong evolution of internal charging with the radiative environment, which has been ascribed to the influence of radiation-induced conductivity. We can notice in these figures that charging kinetics is not constant and strongly depends on the evolution of RIC with the high energy flux and the cumulative radiation dose. Despite high peaks observed on the charging kinetics at the time close to 9×10^6 s, Kapton and, to a lower extent, PEEK present a charging potential that tends to gradually decline in amplitude due to the gradual rise of RIC with cumulative dose. Therefore, this effect tends to smooth the influence of environment dynamics on internal charging. However, steep charging kinetics can be observed for extreme events (as observed for time just before 9×10^6 s) during which RIC (and its increase) is not sufficient to cope with charge implantation. For PEEK, for which RIC evolution is lower than for Kapton, charging levels are higher and we can observe the presence of different charging peaks. This material is more sensitive to environmental variations. Charging peaks are due to the fact that the RIC does not evolve at the same rate as charge implantation and electron current. As described earlier, RIC depends on different physical parameters, such as trapping and recombination, which induce a nonlinearity between RIC and the ionizing current. In other words, if charge implantation and ionizing current are enhanced by a factor of 10, RIC should be increased by a factor lower than 10, leading to a rise of the charging kinetics and charging levels. Potentials reached on PEEK are more critical than for Kapton. ETFE presents a very different charging behavior: this material presents a very

weak induced conductivity. Then, we can observe a constant charge accumulation over time with lower dependence on the ionizing environment. As demonstrated previously in our experimental task, this material should present rapidly strong risks of internal charging and electrostatic discharge in MEO and during EOR.

VI. CONCLUSION

This paper demonstrated the risk of internal charging during EOR on different elements such as Printed Circuit Board. The presence of the floating metallic parts on these elements can further the occurrence of electrostatic discharges that may have a significant effect on the onboard electronic (EMC issues, damages, etc). Risks of internal charging and internal electrostatic discharge should be higher in EOR in comparison with GEO: the reason is that fluxes in EOR are more than one decade higher than the electron fluxes met in GEO (see Paulmier *et al.* [23] for GEO fluxes) especially for the high energies ($E > 1$ MeV) which contribute to internal charging. For materials like those used on PCB that present very low conductivity levels, the fact that the spacecraft moves out of the belts does not yield to a decline of the absolute surface potential. Charging levels on the material—as demonstrated in the paper—keep increasing over the successive cycle in and out of the belts yielding to high risk of discharges at some point.

We have been able to bring into evidence through a numerical modeling using dedicated numerical tools that take into account the evolution of the environment as a function of time and the effect of radiation history on the electric conductivity, that space used polymers behave quite differently regarding a varying GPS-like environment and that internal charging strongly depends on the environment and materials properties. This paper demonstrates as well the importance to take into account radiation history during internal and surface charging assessment.

ACKNOWLEDGMENT

This study has been performed in the frame of the ORESTE research project of ONERA. The authors would like to thank Centre National d'Etudes Spatiales for continuous financial and technical support. K. Ryden and A. Hands from the University of Surrey (U.K.) for providing the on-orbit SURF data.

REFERENCES

- [1] G. L. Wrenn, "Conclusive evidence for internal dielectric charging anomalies on geosynchronous communications spacecraft," *J. Spacecraft Rocket*, vol. 32, no. 3, pp. 514–520, May/Jun. 1995.
- [2] G. L. Wrenn, D. J. Rodgers, and K. A. Ryden, "A solar cycle of spacecraft anomalies due to internal charging," *Annales Geophysicae*, vol. 20, pp. 953–956, Jul. 2002.
- [3] G. L. Wrenn and R. J. K. Smith, "The ESD threat to GEO satellites: Empirical models for observed effects due to both surface and internal charging," in *Proc. ESA Symp. Environ. Modelling Space-Based Appl.*, ESTEC, Noordwijk, NL, SP-392, 1996.
- [4] T. Paulmier, B. Dirassen, M. Arnaout, D. Payan, and N. Balcon, "Radiation-induced conductivity of space used polymers under high energy electron irradiation," *IEEE Trans. Plasma Sci.*, vol. 43, no. 9, pp. 2907–2914, Sep. 2015.
- [5] B. Gross, R. M. Faria, and G. F. Ferreira, "Radiation-induced conductivity in Teflon irradiated by X-rays," *J. Appl. Phys.*, vol. 52, no. 2, pp. 571–578, 1981.
- [6] A. P. Tyutnev, V. S. Saenko, I. A. Smirnov, and E. D. Pozhidaev, "Radiation-induced conductivity in polymers during long-term irradiation," *High Energy Chem.*, vol. 40, no. 5, pp. 319–330, 2006.
- [7] A. Whittlesey and H. B. Garrett, "NASA's technical handbook for avoiding on-orbit ESD anomalies due to internal charging effects," in *Proc. 6th Spacecraft Charging Technol. Conf.*, AFRL-VS-TR-20001578, 1 Sep. 2000.
- [8] G. Wilson, J. R. Dennison, A. E. Jensen, and J. Dekany, "Electron energy-dependent charging effects of multilayered dielectric materials," *IEEE Trans. Plasma Sci.*, vol. 41, no. 12, pp. 3536–3544, Dec. 2013.
- [9] J. J. Likar, K. Hartojo, G. Ott, J. Bird, and N. P. Meredith, "Spacecraft charging related risk of floating connector pins," *IEEE Trans. Plasma Sci.*, vol. 46, no. 1, pp. 201–206, Jan. 2018.
- [10] J.-C. Matéo-Vélez, A. Sicard, D. Payan, N. Ganushkina, N. P. Meredith, and I. Sillanpää, "Spacecraft surface charging induced by severe environments at geosynchronous orbit," *Space Weather*, vol. 16, no. 1, pp. 89–106, Jan. 2018.
- [11] J.-C. Matéo-Vélez *et al.*, "Estimating space environment effects during all-electric telecom satellite missions," in *Proc. 35th Int. Electr. Propuls. Conf.*, Georgia Inst. Technol., Atlanta, GA, USA, Oct. 2017.
- [12] J. I. Vette, *The AE-8 Trapped Electron Model Environment*, document NASA-TM-107820, 1991.
- [13] B. N. S. Rao, "A simple formula for the transmission and absorption of monoenergetic electrons," *Nucl. Instrum. Methods*, vol. 44, pp. 155–156, Sep. 1966.
- [14] *Spacecraft Charging*, document Rec. ECSS-E-ST-20-06C, 2008.
- [15] D. Payan, "Vacuum electrostatic discharges," in *Space Environment: Prevention of Risks Related to Spacecraft Charging*. Toulouse, France: Cepadues Editions, 2002, ch. 12, pp. 317–366.
- [16] R. Coelho, P. Jestin, L. Levy, and D. Sarraill, "On the return-voltage buildup in insulating materials," *IEEE Trans. Electr. Insul.*, vol. EI-22, no. 6, pp. 683–690, Dec. 1987.
- [17] K. A. Ryden and A. D. P. Hands, "Modeling of electric fields inside spacecraft dielectrics using in-orbit charging current data," *IEEE Trans. Plasma Sci.*, vol. 45, no. 8, pp. 1927–1932, Aug. 2017.
- [18] A. Rose, "Recombination processes in insulators and semiconductors," *Phys. Rev. J. Arch. Published Amer. Phys. Soc.*, vol. 97, pp. 322–333, Jan. 1955.
- [19] J. F. Fowler, "X-ray induced conductivity in insulating materials," *Proc. Roy. Soc. London A, Math. Phys. Sci.*, vol. 236, no. 1207, pp. 464–480, 1956.
- [20] T. Paulmier, B. Dirassen, M. Belhaj, and D. Rodgers, "Charging properties of space used dielectric materials," *IEEE Trans. Plasma Sci.*, vol. 43, no. 9, pp. 2894–2900, Sep. 2015.
- [21] T. Paulmier *et al.*, "Aging effect and induced electric phenomena on dielectric materials irradiated with high energy electrons," *IEEE Trans. Plasma Sci.*, vol. 41, no. 12, pp. 3422–3428, Dec. 2013.
- [22] *NASA Technical Handbook: Mitigating In-Space Charging Effects—A Guideline*, document Rec. NASA-HDBK-4002A, Mar. 2011.
- [23] T. Paulmier, A. Sicard-Piet, D. Lazaro, M. Arnaout, and D. Payan, "Analysis of charging kinetics on space dielectrics under representative worst case geostationary conditions," *IEEE Trans. Plasma Sci.*, vol. 43, no. 9, pp. 2849–2855, Sep. 2015.